A Simple Yet Robust Nonlinear Function for Low-Light Image Enhancement Task

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***Abstract*—We present a novel, parameter-free nonlinear transformation for low-light image enhancement that operates directly on individual pixel values. This simple yet powerful function requires no prior knowledge or external tuning, and enhances image brightness and contrast by leveraging only the input image itself. When applied iteratively, the method achieves optimal results after just three applications. Despite its minimalism, our approach outperforms recent state-of-the-art methods on benchmarks. This highlights the potential of simple signal processing operations for emergent enhancement, and suggests directions for theoretical analysis, integration with deep learning, and deployment in real-world vision systems.**

***Index Terms*—Low-light image enhancement, iterative enhancement, parameter-free methods, signal processing, image processing.**

I. Introduction

DIGITAL image processing is essential in fields such as satellite imagery [1], [2], autonomous driving [3], [4], surveillance [5], and medical imaging [6], [7]. A key challenge is low-light image enhancement (LLIE), which mitigates degradations such as low brightness, noise, and color distortions [8], [9]. These degradations impair both human interpretation and algorithmic analysis in critical applications. Traditional methods [10], [11] often fail to preserve fine details and natural appearance [12]. The rise of deep learning has significantly advanced LLIE with techniques leveraging CNNs and Transformers offering superior solutions [13], [14], [15]. However, despite their progress, these approaches remain constrained by model complexity, data dependency, and limited generalizability. This indicates a gap in our fundamental understanding of image structure and its transformation under illumination [16]. This emphasis on complexity has created an implicit belief in the community: that effective enhancement must be complicated. Addressing this gap may lead to more efficient and inherently robust solutions grounded in the intrinsic properties of visual data.

In contrast to prior LLIE methods that rely on deep networks [17], [18], [19], hyperparameter tuning [20], [21], [22], or complex optimization, we present — to the best of our knowledge — the first closed-form, parameter-free, pixel-wise nonlinear function that operates solely on the input image without any training or external priors. Each pixel is enhanced independently through a deterministic function derived solely from the image itself. This minimalist formulation yields notable advantages in efficiency, interpretability, and potential generalizability. Although not explicitly designed for benchmarking, the proposed function achieves strong results, surpassing several recent LLIE methods. This suggests deeper structural principles and shows that even simple transformations can be unexpectedly effective. This invites a reassessment of foundational assumptions in image enhancement.

While a complete theoretical derivation is outside the present scope, the proposed function exhibits structural properties consistent with established perceptual principles. In particular, its adaptive amplification of low-intensity values aligns with Weber's notion of proportional sensitivity, and its saturation for high-intensity values parallels the compression mechanism in Retinex [23]. These correspondences suggest that the function may represent a compact instance of a more general formulation for illumination correction, meriting further analytical study in future work.

The main contributions of this letter are summarized as follows:

- We propose, for the first time, a closed-form, parameter-free, deterministic LLIE function that processes each pixel independently and operates solely on the input image — unlike prior methods requiring training, priors, or hyperparameter tuning.
- Despite its minimal design, our approach surpasses recent state-of-the-art (SOTA) LLIE methods on standard benchmarks, showing that effective enhancement does not require deep or heavily tuned models.
- This finding challenges the prevailing belief in LLIE that complexity is necessary, and suggests that simple, interpretable transformations may embody fundamental principles applicable across image and signal processing.

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II. Methodology

A. Problem Statement and Objective

Let $X \in [0,1]^{m\times n}$ be a normalized low-light image, where pixel values are compressed near zero due to poor illumination, resulting in loss of contrast and detail in shadowed regions.

The objective is to construct a pointwise mapping $f : [0,1] \to [0,1]$ such that the output $Y = f(X)$ enhances visibility in dark areas while preserving mid-tones and avoiding overexposure. To achieve this, f should strongly amplify low intensities, transition smoothly through mid-range values, and saturate near the upper end to protect highlights.

For stability and visual quality, f must be strictly increasing, differentiable, and bounded within $[0,1]$, ensuring that brightness order is preserved, that transitions remain smooth, and that no clipping occurs. The enhancement gain is defined as:

$$\Delta(x) = f(x) - x, \quad x \in X, \tag{1}$$

which quantifies the amount of boost per pixel. Ideally, this gain should peak for moderately dark inputs and diminish near both ends of the range, enabling adaptive enhancement without distortion. In the next section, we design f by shaping this gain function accordingly.

B. Mathematical Formulation

To construct an enhancement function that satisfies the objectives outlined previously, we begin by modeling the enhancement gain $\Delta(x)$. Since the purpose of enhancement is to selectively amplify low-intensity values, the gain should be positive and maximal in the lower range of the domain, gradually decreasing to zero as $x \to 1$. This suggests that $\Delta(x)$ should be defined by a nonnegative function that is unimodal over $[0,1]$ and vanishes at both endpoints.

A natural hypothesis is to model $\Delta(x)$ as a product of terms that enforce both the desired shape and boundary conditions. To this end, we consider the form

$$\Delta(x) = \alpha(1-x)(x-x^2), \tag{2}$$

where $\alpha \in [0,1]$ is a tunable parameter that modulates the enhancement strength. This expression satisfies the basic requirements: it vanishes at $x = 0$ and $x = 1$, is strictly positive in $(0,1)$, and attains a maximum at an interior point. The component $x - x^2 = x(1-x)$ gives the gain a bell-shaped structure, while the leading $(1-x)$ factor introduces additional emphasis on the dark end of the range.

The corresponding enhancement function is then given by

$$f(x) = x + \alpha(1-x)(x-x^2). \tag{3}$$

To validate that this formulation produces the intended behavior, we expand the expression algebraically:

$$\begin{aligned} f(x) &= x + \alpha(1-x)(x-x^2) \\ &= x + \alpha(x - 2x^2 + x^3). \end{aligned} \tag{4}$$

By setting $\alpha = 1$, which corresponds to the case of full-strength enhancement, we obtain the final polynomial form:

$$f(x) = 2x - 2x^2 + x^3. \tag{5}$$

This function is a cubic polynomial that maps the unit interval to itself. It is strictly increasing, smooth, and bounded, satisfying all the conditions set forth in the formulation of the enhancement objective.

The critical points of Δ satisfy:

$$\frac{d\Delta}{dx} = 1 - 4x + 3x^2 = 0 \quad \Longrightarrow \quad x = \frac{4 \pm \sqrt{16-12}}{6} = \frac{1}{3},\ 1. \tag{6}$$

Thus, $f(x)$ yields a gain function $\Delta(x)$ whose properties align with perceptual principles: (i) for $x \ll 1$, the nearly constant slope reflects Weber's law of proportional sensitivity, supporting strong but proportional amplification in dark regions; (ii) the peak at $x = \frac{1}{3}$ indicates the intensity level where enhancement is maximized, consistent with mid-level sensitivity in human vision; and (iii) as $x \to 1$, $\Delta(x) \to 0$, paralleling response compression in Retinex and human adaptation models that prevent highlight saturation.

Evaluating $\Delta(.)$ at $x = \frac{1}{3}$ yields the global maximum

$$\max_{x\in[0,1]} \Delta(x) = \Delta\left(\frac{1}{3}\right) = \frac{1}{3} - \frac{2}{9} + \frac{1}{27} = \frac{4}{27} \approx 0.148. \tag{7}$$

Since no pixel can increase by more than 0.148 in a single application, one iteration provides only a modest improvement. For instance, $f(0.1) = 0.181$ so $\Delta(0.1) = 0.081$, and $f(0.5) = 0.625$ so $\Delta(0.5) = 0.125$. Consequently, iterative application of f is required to accumulate these incremental gains and achieve satisfactory brightness in lowlight images.

C. Iterative Enhancement

The iterative process is described as:

$$x_{(k)} = f(x_{(k-1)}), \quad k = 1, 2, \ldots, K, \tag{8}$$

where K denotes the number of enhancement iterations, and $x_{(0)} = x$ refers to the initial pixel intensity. The strategy is to leverage the original input $M = -x_{(0)}$ as the modulation anchor, which naturally leads to the recursive map:

$$x_k = x_{k-1} + (1+M)(x_{k-1} - x_{k-1}^2). \tag{9}$$

Considering the case where M is dynamic $M_{k-1} = -x_{k-1}$, the enhancement quickly weakens as pixels brighten, leading to premature stagnation in underexposed areas. Fixing $M = -x_{(0)}$ anchors the enhancement to the original image, ensuring consistent correction throughout the iteration.

Fig. 1(a) shows the overall pipeline of the proposed method, while Fig. 1(b) illustrates the enhancement function across iterations under fixed modulation. The function remains monotonic and exhibits increasingly strong amplification for low-intensity inputs, particularly within the first three iterations. This monotonic behavior confirms that pixel intensities are consistently elevated without reversals, preserving local contrast. Beyond three iterations, the curves begin to saturate, and the monotonicity of updates is no longer guaranteed across the full range.

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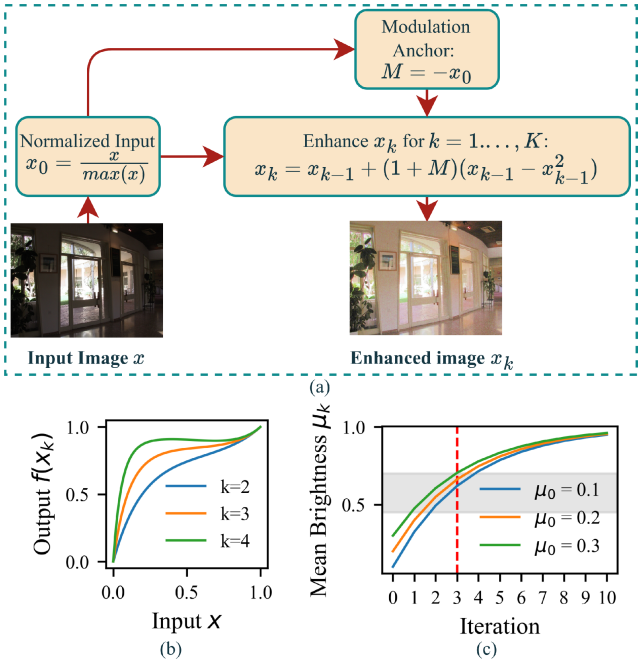

Fig. 1. Iterative image enhancement: (a) general pipeline with modulation anchor (applied to the V channel when using HSV color space), (b) output functions for different iterations, and (c) convergence analysis showing brightness evolution with gray band as convergence area and red line marking iteration to convergence.

D. Perceptual Convergence Criteria

The enhancement process is defined as repeated application of the nonlinear mapping f. While the iteration converges mathematically to full brightness as $k \to \infty$, such a limit is undesirable since excessive iterations cause overexposure and loss of detail. We therefore redefine convergence in perceptual terms using the global mean brightness of the image. Let $X^{(k)} \in [0,1]^{m \times n}$ denote the image after k iterations, where (i,j) indexes pixel position. Its mean brightness is

$$\mu_k = \frac{1}{mn} \sum_{i,j} X_{i,j}^{(k)}. \tag{10}$$

The image is said to be perceptually converged if

$$\mu_k \in [\tau_1, \tau_2], \quad [\tau_1, \tau_2] = [0.45, 0.70], \quad \forall \mu_0 \leq \mu_L, \tag{11}$$

where μ_0 is the initial mean and μ_L is the mean of a low-light image (typically $\mu_L \leq 0.30$ for most low-light images). Since μ_k may both enter and leave the interval, we define the perceptual convergence point as the maximum iteration satisfying (11).

For a dark image with $\mu_0 < \tau_1$, the mean after k steps can be approximated as

$$\mu_k \approx \mu_0 + k \cdot \delta_{\text{avg}}, \tag{12}$$

where δ_{avg} is the average per-pixel gain per iteration. Solving (12) for k to satisfy (11) gives

$$k \geq \frac{\tau_1 - \mu_0}{\delta_{\text{avg}}}, \quad k \leq \frac{\tau_2 - \mu_0}{\delta_{\text{avg}}}. \tag{13}$$

With $\delta_{\text{avg}} \approx 0.12$, only $k = 3$ satisfies the convergence condition. While $k = 2$ may enter the target band, it fails the maximum-iteration rule, and $k = 4$ overshoots the upper bound. As shown in Fig. 1(c), $k = 3$ alone remains valid across different μ_0, providing balanced enhancement and directly illustrating the influence of varying k in (12)–(13).

TABLE I
QUANTITATIVE COMPARISON ACROSS DIFFERENT DATASETS USING DIFFERENT VALUES OF k IN HSV AND RGB COLOR SPACES

Dataset	Metrics	HSV2	HSV3	HSV4	RGB2	RGB3	RGB4
LIME	NIQE↓	3.80	3.64	**3.58**	3.78	3.71	3.77
	BRISQUE↓	16.35	**15.11**	15.23	17.12	16.23	16.41
	PIQE↓	**31.83**	32.51	32.59	32.99	34.36	36.80
LOLv1	PSNR↑	17.93	**19.45**	18.90	18.04	19.28	16.68
	MSE↓	167	150	97	162	144	**90**
	PI↓	0.52	**0.33**	0.36	0.50	0.34	0.42

III. EXPERIMENTS AND RESULTS

A. Experimental Setup

We evaluate our method on standard low-light datasets. DICM [24], LIME [21], MEF [25], NPE [26], and VV[1] are combined into an **unpaired set (U)**, while LOLv1 [27], LOLv2 [28] form a **paired set (P)**. We compare against SNRA [29], UTVNet [30], RetinexFormer (RFormer) [31], LLFormer [32], and LLDiffusion (LLDiff.) [33]. Evaluation uses PSNR, MSE, Perceptual Index (PI) [34], BRISQUE, Natural Image Quality Evaluator (NIQE) [35], and Perception-based Image Quality Evaluator (PIQE) [36], which together assess perceptual quality and distortion.

Before applying the enhancement, a simple normalization step ensures consistent dynamic range $x = x/\max(x)$.

B. Color Space and Iteration Analysis

We examine the influence of both color space and iteration count on the quality of image enhancement. The notation *HSV2*, *HSV3*, and *HSV4* refers to enhancement performed in the HSV color space with 2, 3, and 4 iterations, respectively; similarly, *RGB2*, *RGB3*, and *RGB4* denote the corresponding processes in the RGB color space. In all subsequent tables, the **best** performance scores are indicated in bold, while the second-best scores are underlined. This naming convention and formatting scheme are consistently applied throughout the following discussion.

Table I shows that HSV consistently outperforms RGB across datasets. On LOLv1, HSV3 achieves the highest PSNR, lowest PI, and competitive MSE, while RGB4, despite its lowest MSE, suffers in PSNR and PI, indicating over-enhancement. RGB3 fares slightly better but still trails HSV3. Qualitative results in Fig. 2 confirm that HSV3 preserves natural color and balanced brightness, whereas higher-iteration RGB tends to cause hue shifts and overexposure. From a human visual perception perspective, the superiority of HSV is consistent with the fact that the H (hue), S (saturation), and V (value) components separate chromatic and luminance information. Enhancing only the V

[1] https://sites.google.com/site/vonikakis/datasets

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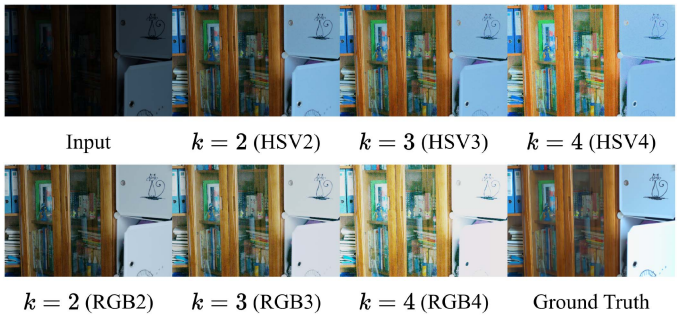

Fig. 2. Comparison of enhancement results on a sample image from the LOLv1 dataset. Top row: Input image followed by HSV enhancement with $k = 2, 3, 4$. Bottom row: RGB enhancement with $k = 2, 3, 4$, followed by the ground truth.

TABLE II
QUANTITATIVE COMPARISON ON GROUPED DATASETS

Dataset	Metrics	SNRA	UTVNet	RFormer	LLFormer	LLDiff.	HSV3
U	NIQE↓	3.75	3.75	3.88	3.67	7.06	**3.47**
	BRISQUE↓	23.52	14.59	14.76	14.60	20.40	**14.58**
	PIQE↓	41.58	30.61	37.23	35.15	40.50	**30.53**
P	PSNR↑	17.82	18.57	20.21	16.29	**24.40**	20.52
	MSE↓	150	175	**143**	182	151	147
	PI↓	0.40	0.47	**0.32**	0.62	0.42	**0.32**

U = unpaired (DICM/LIME/MEF/VV/NPE), P = paired (LOLv1/LOLv2-real/LOLv2-synthetic). One checkpoint per baseline: rformer (LOLv2-synthetic), LLFormer (FIVEK/SSIM), LLDiffusion (LOL).

channel aligns with how the human visual system processes brightness largely independently of color, reducing the risk of hue distortion. In contrast, direct RGB manipulation alters all three color primaries simultaneously, which can disrupt the relative cone responses and lead to unnatural color balance.

C. Results and Analysis

Table II and Fig. 3 summarize the quantitative and qualitative comparisons. Recent transformer- and diffusion-based methods (RetinexFormer, LLFormer, LLDiffusion) release multiple checkpoints tuned for different datasets or metrics (e.g., separate best-PSNR and best-SSIM models). Although this practice inflates reported results, it does not reflect real-world use, where a single model must generalize. For fairness, we evaluate one checkpoint per method across all benchmarks.

RetinexFormer, when trained and evaluated on LOLv1, achieves high PSNR (25.87) on that dataset but performs much worse when trained on LOLv2-synthetic (as reported in Table II) and deteriorates further on unpaired datasets, revealing overfitting. LLFormer shows similar behavior: its 'LOL best PSNR' checkpoint achieves 25.41 PSNR on LOL, but its unpaired metrics (e.g., NIQE 3.73, BRISQUE 16.03, PIQE 31.21) are poor. LLDiffusion, likewise, excels on LOL but fails to generalize beyond it. These results confirm that checkpoint selection masks generalization issues, whereas HSV3 achieves consistently strong results across both U and P without dataset-specific tuning. Additional baselines SCI [37], RUAS [38], and KinD++ [39] yielded weaker results (PSNR $\leq$ 18.0, NIQE $\geq$ 3.8, PI $\geq$ 0.43) than HSV3 and are omitted from Table II for clarity.

Fig. 3. Qualitative comparison on DICM and NPE. Only a subset of baselines is shown (Rformer, LLFormer, LLDiff., UTVNet) alongside HSV3 for clarity.

The qualitative results reinforce these trends: RetinexFormer suffers from color casts, LLDiffusion introduces banding, UTVNet produces yellowish bias, and LLFormer shows underexposure. In contrast, HSV3 maintains natural colors, smooth skies, and balanced illumination, offering stable visual quality across diverse datasets. Overall, HSV3 combines simplicity with robustness, delivering balanced metrics and superior perceptual results. A limitation is that in extremely noisy conditions the enhancement may amplify noise, motivating future integration with denoising.

IV. CONCLUSION

We introduced a simple, parameter-free nonlinear function for LLIE that operates directly on pixel values without training or tuning. Despite its minimalism, the method outperforms several SOTA approaches in both visual quality and objective metrics, with three iterations yielding optimal results.

This work highlights the power of lightweight, learning-free methods in LLIE. Its simplicity not only makes it practical but also opens the door to further exploration in theory, integration with learning models, and real-world deployment. However, the method addresses illumination only and does not explicitly remove noise. In extremely low-light scenes with substantial sensor noise, this can result in amplified noise patterns and reduced perceptual quality. Future work will focus on integrating the enhancement function with denoising, particularly for challenging imaging environments.

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REFERENCES

[1] M. Bakirci, "Advanced ship detection and ocean monitoring with satellite imagery and deep learning for marine science applications," *Regional Stud. Mar. Sci.*, vol. 81, 2025, Art. no. 103975. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S235248552400608X

[2] T.-A. Bui, P.-J. Lee, C.-S. Liang, P.-H. Hsu, S.-H. Shiu, and C.-K. Tsai, "Edge-computing-enabled deep learning approach for low-light satellite image enhancement," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 4071–4083, 2024.

[3] K. Xu, Y. Zhang, J. Li, X. Cheng, and Z. Liu, "A zero-reference night-time road image enhancement method," *Proc. SPIE*, vol. 13521, 2025, Art. no. 135210A, doi: 10.1117/12.3057936.

[4] C. Qian et al., "Allweather-Net: Unified image enhancement for autonomous driving under adverse weather and low-light conditions," in *Proc. Int. Conf. Pattern Recognit.*, 2025, pp. 151–166.

[5] Q. Du, S. Zhang, and S. Yang, "BLP-YOLOv10: Efficient safety helmet detection for low-light mining," *J. Real-Time Image Process.*, vol. 22, no. 1, 2025, Art. no. 10.

[6] W. Birkfellner, *Applied medical image processing: A basic course*. Boca Raton, FL, USA: CRC Press, 2024.

[7] E. S. Mira et al., "Early diagnosis of oral cancer using image processing and artificial intelligence," *Fusion: Pract. Appl.*, vol. 14, no. 1, pp. 1–2, 2024.

[8] G. Halford, A. C. D. II, and C. P. Bailey, "Brightening the shadows: Low-light image feature enhancement," *Proc. SPIE*, vol. 13033, 2024, Art. no. 1303303, doi: 10.1117/12.3013958.

[9] Y. Wu et al., "Towards a flexible semantic guided model for single image enhancement and restoration," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 12, pp. 9921–9939, Dec. 2024.

[10] K. Singh and A. S. Parihar, "Illumination estimation for nature preserving low-light image enhancement," *Vis. Comput.*, vol. 40, no. 1, pp. 121–136, 2024.

[11] S. Hao, X. Han, Y. Guo, X. Xu, and M. Wang, "Low-light image enhancement with semi-decoupled decomposition," *IEEE Trans. Multimedia*, vol. 22, pp. 3025–3038, 2020.

[12] H. Zhu, Y. Zhao, R. Wang, R. Wang, W. Chen, and X. Gao, "LLISP: Low-light image signal processing net via two-stage network," *IEEE Access*, vol. 9, pp. 16736–16745, 2021.

[13] G. Tang, J. Ni, Y. Chen, W. Cao, and S. X. Yang, "An improved cycleGAN-based model for low-light image enhancement," *IEEE Sensors J.*, vol. 24, no. 14, pp. 21879–21892, Jul. 2024.

[14] X. Yi, H. Xu, H. Zhang, L. Tang, and J. Ma, "Diff-retinex++ : Retinex-driven reinforced diffusion model for low-light image enhancement," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 8, pp. 6823–6841, Aug. 2025.

[15] S. Wang, Q. Lu, B. Peng, Y. Nie, and Q. Tao, "Dpec: Dual-path error compensation for low-light image enhancement," *Neurocomputing*, vol. 637, 2025, Art. no. 129980. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0925231225006526

[16] J. Wang, S. Huang, Z. Huo, S. Zhao, and Y. Qiao, "Bilateral enhancement network with signal-to-noise ratio fusion for lightweight generalizable low-light image enhancement," *Sci. Rep.*, vol. 14, no. 1, Nov. 2024, Art. no. 29832.

[17] L. Xu, C. Hu, Y. Hu, X. Jing, Z. Cai, and X. Lu, "UPT-flow: Multi-scale transformer-guided normalizing flow for low-light image enhancement," *Pattern Recognit.*, vol. 158, 2025, Art. no. 111076. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0031320324008276

[18] Y. Wen, P. Xu, Z. Li, and W. X. ATO, "An illumination-guided dual attention vision transformer for low-light image enhancement," *Pattern Recognit.*, vol. 158, 2025, Art. no. 111033.

[19] S. Jiang, Y. Mei, P. Wang, and Q. Liu, "Exposure difference network for low-light image enhancement," *Pattern Recognit.*, vol. 156, 2024, Art. no. 110796. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0031320324005478

[20] M. Li, J. Liu, W. Yang, X. Sun, and Z. Guo, "Structure-revealing low-light image enhancement via robust retinex model," *IEEE Trans. Image Process.*, vol. 27, no. 6, pp. 2828–2841, Jun. 2018.

[21] X. Guo, Y. Li, and H. Ling, "LIME: Low-light image enhancement via illumination map estimation," *IEEE Trans. Image Process.*, vol. 26, no. 2, pp. 982–993, Feb. 2017.

[22] Y. Ren, Z. Ying, T. H. Li, and G. Li, "LECARM: Low-light image enhancement using the camera response model," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 29, no. 4, pp. 968–981, Apr. 2019.

[23] E. H. Land and J. J. McCann, "Lightness and retinex theory," *J. Opt. Soc. Amer.*, vol. 61, no. 1, pp. 1–11, 1971.

[24] C. Lee, C. Lee, and C.-S. Kim, "Contrast enhancement based on layered difference representation," in *Proc. 19th IEEE Int. Conf. Image Process.*, 2012, pp. 965–968.

[25] K. Ma, K. Zeng, and Z. Wang, "Perceptual quality assessment for multi-exposure image fusion," *IEEE Trans. Image Process.*, vol. 24, no. 11, pp. 3345–3356, Nov. 2015.

[26] S. Wang, J. Zheng, H.-M. Hu, and B. Li, "Naturalness preserved enhancement algorithm for non-uniform illumination images," *IEEE Trans. Image Process.*, vol. 22, no. 9, pp. 3538–3548, Sep. 2013.

[27] C. Wei, W. Wang, W. Yang, and J. Liu, "Deep retinex decomposition for low-light enhancement," 2018, *arXiv:1808.04560*.

[28] W. Yang, W. Wang, H. Huang, S. Wang, and J. Liu, "Sparse gradient regularized deep retinex network for robust low-light image enhancement," *IEEE Trans. Image Process.*, vol. 30, pp. 2072–2086, 2021.

[29] X. Xu, R. Wang, C.-W. Fu, and J. Jia, "SNR-aware low-light image enhancement," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 17693–17703.

[30] C. Zheng, D. Shi, and W. Shi, "Adaptive unfolding total variation network for low-light image enhancement," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2021, pp. 4439–4448.

[31] Y. Cai, H. Bian, J. Lin, H. Wang, R. Timofte, and Y. Zhang, "Retinexformer: One-stage retinex-based transformer for low-light image enhancement," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2023, pp. 12470–12479.

[32] T. Wang, K. Zhang, T. Shen, W. Luo, B. Stenger, and T. Lu, "Ultra-high-definition low-light image enhancement: A benchmark and transformer-based method," in *Proc. AAAI Conf. Artif. Intell.*, 2023, vol. 37, no. 3, pp. 2654–2662.

[33] T. Wang et al., "LLDiffusion: Learning degradation representations in diffusion models for low-light image enhancement," *Pattern Recognit.*, vol. 166, 2025, Art. no. 111628. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0031320325002882

[34] Y. Blau and T. Michaeli, "The perception-distortion tradeoff," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 6228–6237.

[35] A. Mittal, R. Soundararajan, and A. C. Bovik, "Making a 'completely blind' image quality analyzer," *IEEE Signal Process. Lett.*, vol. 20, no. 3, pp. 209–212, Mar. 2013.

[36] N. Venkatanath, D. Praneeth, M. C. Bh, S. S. Channappayya, and S. S. Medasani, "Image quality evaluation using perception based features," in *Proc. 21st Nat. Conf. Commun.*, 2015, pp. 1–6.

[37] L. Ma, T. Ma, R. Liu, X. Fan, and Z. Luo, "Toward fast, flexible, and robust low-light image enhancement," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 5627–5636.

[38] R. Liu, L. Ma, J. Zhang, X. Fan, and Z. Luo, "Retinex-inspired unrolling with cooperative prior architecture search for low-light image enhancement," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 10556–10565.

[39] Y. Zhang, X. Guo, J. Ma, W. Liu, and J. Zhang, "Beyond brightening low-light images," *Int. J. Comput. Vis.*, vol. 129, pp. 1013–1037, 2021.

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